Chapter 3: Discrete Random Variables

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- To analyze random experiments, we focus on numerical aspects of the experiment.
- For example, if an entire soccer game is a random experiment, then numerical results like #goals, #shots, #fouls, etc. are *random variables*.

Example

We toss a coin 3 times. Then $S = \{HHH, HHT, \ldots, TTT\}$. We are interested in the number of heads: Define a random variable X by $X(HHH) = 3$, $X(HHT) = 2$,... So $X \in \{0, 1, 2, 3\}$. Note that X is a function from S to $\mathbb{R}$ with range $\{0, 1, 2, 3\}$.

A **random variable** is a function $X : S \to \mathbb{R}$. Its range is denoted by Range(X) or R_X. Random variables are denoted by X, Y, Z, etc.

Examples of random variables

- I toss a coin until heads comes up. Let X be the total number of tosses (including the last toss). $R_X = \mathbb{N}$.
- We record the wind velocity $V = (V_1, V_2)$ at a particular time and place. Let X be the wind speed: $X = \sqrt{V_1^2 + V_2^2}$. So $R_X = [0, \infty)$. Let Y be the wind direction: $Y = (360/2\pi)\text{atan}(Y_2/Y_1)$.[1] Then $R_Y = [0, 360)$.
- I roll two dice and let Z be the sum of the result. Then $R_Z = \{2, 3, \ldots, 12\}$.
- I ask a random VU-student if she studies AI and encode the answer by $X = 0$ if "no" and $X = 1$ if "yes". Then $R_X = \{0, 1\}$.

[1] The direction is here taken to be relative to East.

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A random variable is **discrete** if its range is countable (i.e. either finite of countably infinite).

Let S be a sample space, P a probability measure on S, and $X\colon S \to \mathbb{R}$ a random variable.

We want to assign probabilities to values in the range $R_X = \{x_1, x_2, \dots\}$ of X.

Idea: For $x_k \in R_X$ we take the probability of the event $A = \{s \in S : X(s) = x_k\}$. We denote this event by $\{X = x_k\}$.[2]

[2] Another notation is $X^{-1}(x_k)$, i.e. the pre-image of x_k under the function X.

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Example

Suppose we roll a die. Then $S = \{1, 2, 3, 4, 5, 6\}$. Define the random variable $X : S \to \mathbb{R}$ by

$$X(s) = \begin{cases} 0 & \text{if } s \text{ is odd} \\ 1 & \text{if } s \text{ is even.} \end{cases}$$

Then

$$P(X = 0) \equiv {}^{3}P(X^{-1}(0)) = P(\{1, 3, 5\}) = 1/2$$

and

$$P(X = 1) \equiv P(X^{-1}(1)) = P(\{2, 4, 6\}) = 1/2.$$

[3] The symbol $\equiv$ means "per definition. Another symbol for this is $:=$.

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Let X be a random variable with range $R_X = \{x_1, x_2, \dots\}$. The function

$$P_X(x_k) = P(\{X = x_k\})^a$$

for $k = 1, 2, \dots$ is called the **probability mass function** (PMF) of X.

[a] Below we will simply write $P(X = x_k)$.

Comment: It is convenient to extent P_X to the real numbers $\mathbb{R}$:

$$P_X(x_k) = \begin{cases} P(X = x) & \text{if } x \in R_X \\ 0 & \text{otherwise} \end{cases}$$

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Example (unfair coin)

We have an unfair coin with $P(H) = p$ for certain $0 < p < 1$. We toss it repeatedly until heads comes up for the first time. Let Y be the total number of coin tosses (including the last toss). Find the PMF of Y.

Solution: Note that $R_Y = \mathbb{N}$ and $S = \{H, TH, TTH, \dots\}$. Now:

$$\begin{aligned} P_Y(1) &= P(Y=1) = P(H) = p \\ P_Y(2) &= P(Y=2) = P(TH) = (1-p)p \\ P_Y(3) &= P(Y=3) = P(TTH) = (1-p)^2 p \\ &\vdots \end{aligned}$$

So

$$P_Y(k) = \begin{cases} (1-p)^{k-1}p & \text{for } k \in \mathbb{N} \\ 0 & \text{otherwise} \end{cases}$$

Properties of the PMF

- $0 \leq P_X(x) \leq 1$ for all x (this is because $P_X(x) = P(\{X = x\})$ and P is a probability measure).
- $\sum_{x \in R_X} P_X(x) = 1$
- for any set $A \subset R_X, P(X \in A) = \sum_{x \in A} P_X(x)$.

The last two statements follow from the fact that $\{\{X = x\} : x \in R_X\}$ form a partition of S. For instance:

$$\begin{aligned}\sum_{x \in R_X} P_X(x) &= \sum_{x \in R_X} P(\{X = x\}) \\ &= P(\cup_{x \in R_x} \{X = x\}) \\ &= P(S) \\ &= 1.\end{aligned}$$

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Example

Probability Mass Function (PMF)

I roll a fair die repeatedly until a number larger than 4 is observed. Let N be the total number of times that I roll the die. Find the PMF of N.

Solution: The sample space consists of all sequences of numbers from the set $\{1, 2, 3, 4, 5, 6\}$ for which the last number is > 4 and all other numbers are ≤ 4. The random variable $N\colon S \to \mathbb{R}$ assigns to each outcome its length and so $R_N = \mathbb{N}$. Furthermore:

$$P_N(1) = 1/3$$
$$P_N(2) = 2/3 \times 1/3$$
$$P_N(3) = (2/3)^2 \times 1/3$$

and so on, hence we conclude that

$$P_N(k) = \begin{cases} (2/3)^{k-1} \times 1/3 & \text{for } k \in \mathbb{N} \\ 0 & \text{otherwise} \end{cases}$$

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- The following is not made explicit in the book, but it will be illuminating to make it explicit.
- For two events $A, B \subset S$ we have defined $P(A \cap B)$, $P(A \cup B)$, and $P(A|B)$.
- Let $X, Y : S \to \mathbb{R}$ be discrete random variables on the same sample space and let $A \subset R_X$ and $B \subset R_Y$.
- We then define $P(X \in A \text{ and } Y \in B)$, $P(Y \in A \text{ or } Y \in B)$, and $P(X \in A | Y \in B)$ by taking pre-images:

$$P(X \in A \text{ and } Y \in B) = P(X^{-1}(A) \cap Y^{-1}(B))$$
$$P(X \in A \text{ or } Y \in B) = P(X^{-1}(A) \cup Y^{-1}(B))$$
$$P(X \in A | Y \in B) = P(X^{-1}(A) | Y^{-1}(B))$$

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Two discrete random variables X and Y are **independent** if

$$P(X = x, Y = y) = P(X = x)P(Y = y)$$

for all $x \in R_X$ and $y \in R_Y$.[a]

[a] The comma in $P(X \in A, Y \in B)$ means "and".

- This is equivalent to independence of the events $X^{-1}(x)$ and $Y^{-1}(y)$ for all $x \in R_X$ and $y \in R_Y$.
- Independence implies that $P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$ for all $A \subset R_X$ and $B \subset R_Y$.
- X and Y are independent if and only if $P(X \in A | Y \in B) = P(X \in A)$ for all $A \subset R_X$ and $B \subset R_Y$ with $P(B) > 0$.
- Intuitively: Knowing that $Y \in B$ does not change the probability that $X \in A$ (and *vice versa*).

Example

The assumption of independence of two random variables can often be justified by knowledge of the system under study.

Example

We toss a coin twice and let X be the number of heads. We then toss the coin two more times and let Y be the number of heads. Calculate $P(X < 2, Y > 1)$.

Solution: We assume that the first two tosses are independent from the last two tosses and therefore

$$\begin{aligned} P(X < 2, Y > 1) &= P(X < 2)P(Y > 1) \\ &= (1 - P(X = 2))P(Y = 2) \\ &= (1 - 1/4) \times 1/4 \\ &= 3/16. \end{aligned}$$

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Independence of more than two random variables

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Discrete random variables $X_1, \ldots, X_n$ are **independent** if

$$P(X_1 = x_1, \ldots, X_n = x_n) = P(X_1 = x_1) \cdots P(X_n = x_n)$$

for all $x_i \in R_{X_i}$.

Example (XOR gate)

Let X_1 and X_2 be independent random bits with $P(X_1 = 1) = P(X_2 = 1) = 1/2$ and define a random bit X_3 by

$$X_3 = \begin{cases} 0 & \text{if } (X_1, X_2) = (0,0) \text{ or } (X_1, X_2) = (1,1) \\ 1 & \text{if } (X_1, X_2) = (1,0) \text{ or } (X_1, X_2) = (0,1) \end{cases}$$

Then $P(X_1 = i, X_3 = j) = 1/4 = P(X_1 = i)P(X_3 = j)$ for all $i, j \in \{0, 1\}$ so X_1 and X_3 are independent and, similarly, X_2 and X_3 are independent. But $P(X_1 = 1, X_2 = 1, X_3 = 1) = 0$, whereas $P(X_1 = 1)P(X_2 = 1)P(X_3 = 1) = 1/8 \neq 0$ so X_1, X_2, X_3 are *not* independent.

Exercises

Section 3.3: Exercises 1, 2, 3, 4, 5

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Special Distributions

- A number of distributions appear in many applications.
- Each of these is associated with a random experiment.
- We will discuss the following distributions: **Bernoulli, Binomial, Negative Binomial (Pascal), Poisson.**

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The Bernoulli Distribution

- We perform an experiment with two possible outcomes[4]: either event A occurs or A does not occur (**Bernoulli experiment**)
- Suppose that $P(A) = p$ for certain $0 \leq p \leq 1$.
- With this experiment we can associate the random variable

$$X = \begin{cases} X = 1 & \text{if } A \text{ occurs} \\ X = 0 & \text{otherwise} \end{cases}$$

- $P(X = 1) = p$ and $P(X = 0) = 1 - p$.

A random variable X has a **Bernoulli distribution** with parameter $0 \leq p \leq 1$ if

$$P_X(x) = \begin{cases} p & \text{for } x = 1 \\ 1 - p & \text{for } x = 0 \end{cases}$$

Notation: $X \sim \textit{Bernoulli}(p)$

[4] Commonly the outcomes are referred to as "success" (A) and "failure" (not A).

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The Bernoulli Distribution

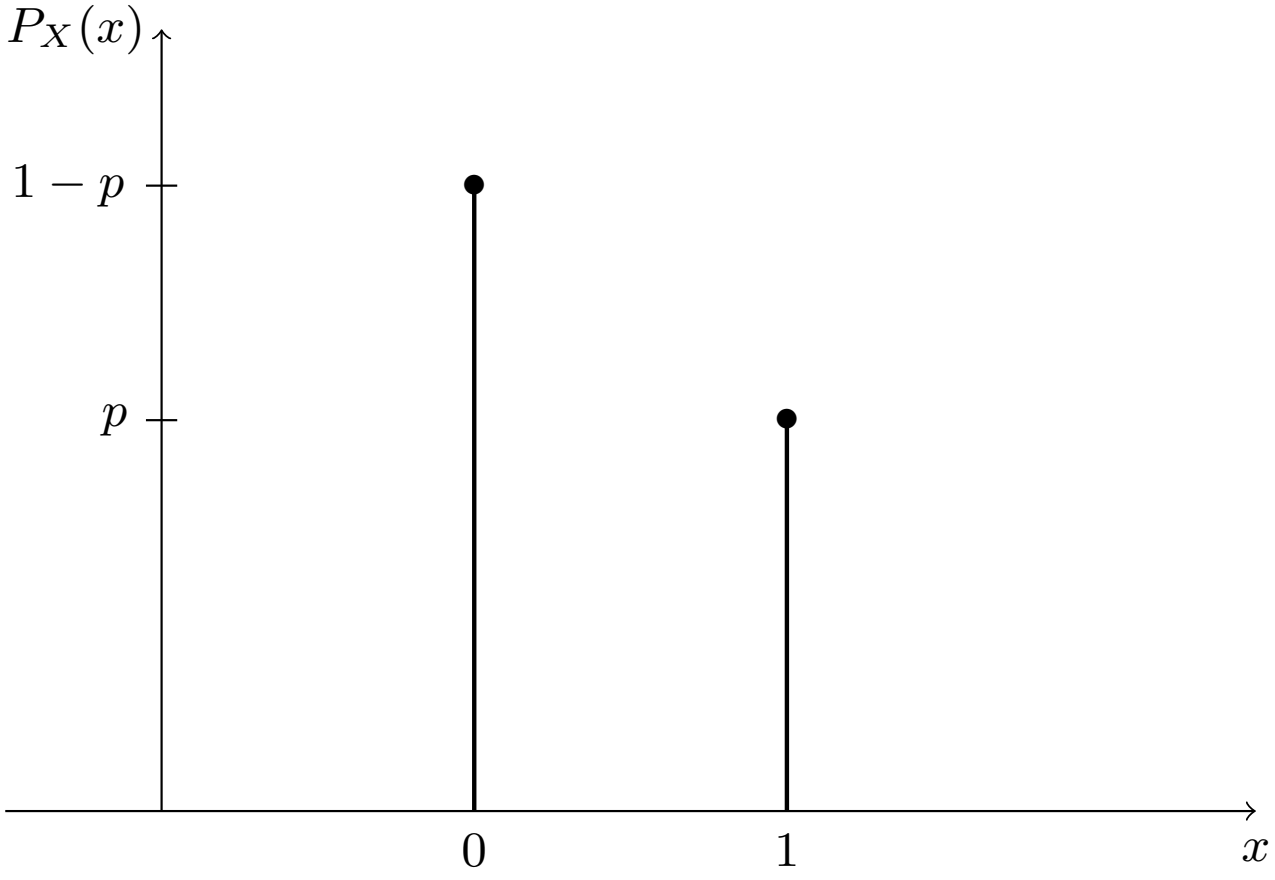

Figure: PMF of a $Bernoulli(p)$ random variable

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The Binomial Distribution

- Suppose we carry out n independent Bernoulli experiments, each with success probability p.
- Let $X \in \{0, 1, 2, \ldots, n\}$ be the total number of successes.
- What is the PMF of X?

$$P_X(0) = (1-p)^n$$

$$P_X(1) = np(1-p)^{n-1} = \binom{n}{1} p(1-p)^{n-1}$$

$$P_X(2) = \binom{n}{2} p^2 (1-p)^{n-2}$$

$$\vdots$$

$$P_X(n) = p^n = \binom{n}{n} p^n$$

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A random variable X has a **Binomial distribution** with parameters n and p if

$$P_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \text{for } k = 0, 1, 2, ..., n.$$

and $P_X(k) = 0$ otherwise. Notation: $X \sim \textit{Binomial}(n, p)$.

- $X \sim \textit{Binomial}(1, p)$ means that $X \sim \textit{Bernoulli}(p)$.
- Let $X \sim \textit{Binomial}(n, p)$ and let $X_1, \ldots, X_n \sim \textit{Bernoulli}(p)$ be independent, then $X = X_1 + \cdots + X_n$.

The Binomial Distribution

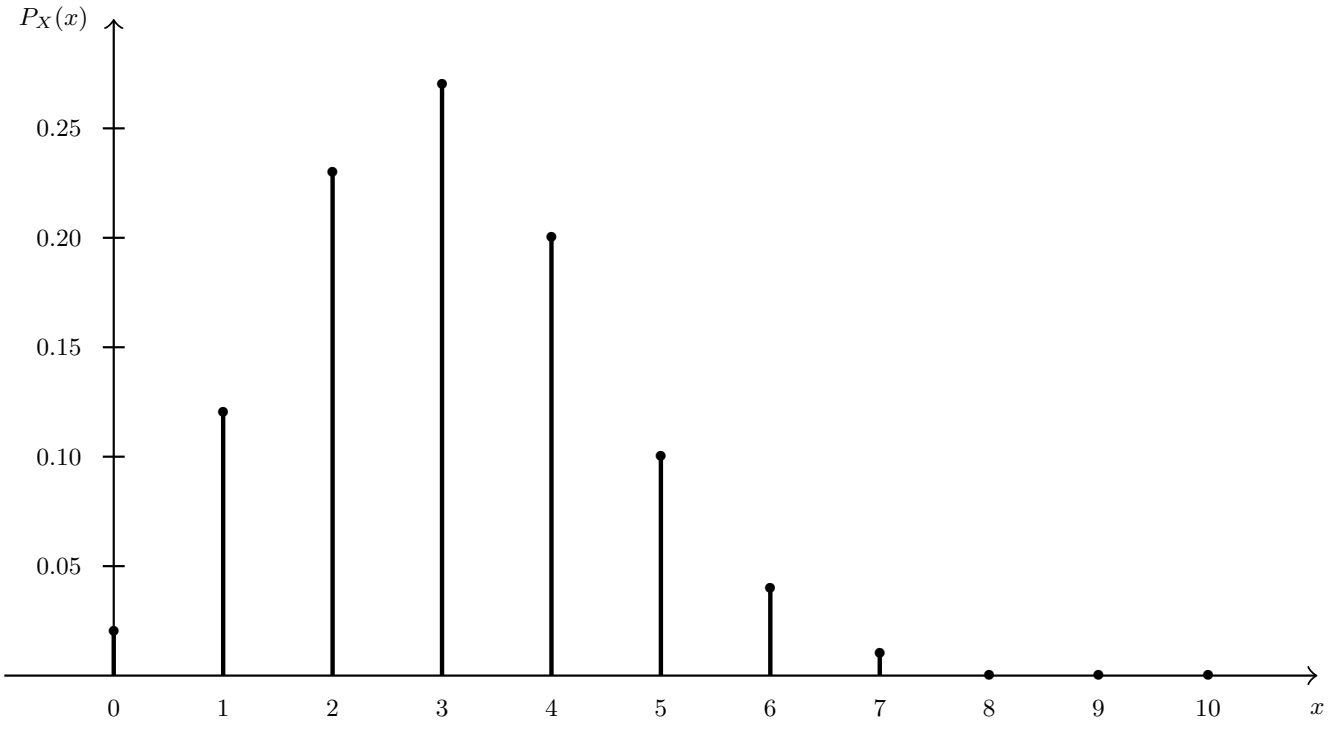

Figure: PMF of a $Binomial(10, 0.3)$ random variable

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Example (sum of independent binomial random variables)

Let $X \sim$ *Binomial*(n, p) and $Y \sim$ *Binomial*(m, p) be independent. Determine the PMF of $Z = X + Y$.

Solution (direct calculation):

$$
\begin{aligned}
P_Z(k) = P(Z = k) &= P(X + Y = k) \\
&= \sum_{i=1}^{n} P(X + Y = k \mid X = i) P(X = i) \\
&= \sum_{i=1}^{n} P(Y = k - i \mid X = i) P(X = i) \\
&= \sum_{i=1}^{n} P(Y = k - i \mid X = i) P(X = i) \\
&= \sum_{i=1}^{n} P(Y = k - i) P(X = i) = \sum_{i=1}^{n} P_X(i) P_Y(k - i).
\end{aligned}
$$

The Binomial Distribution

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Example (continued)

$$\begin{aligned}
\sum_{i=1}^{n} P_X(i)P_Y(k-i) &= \sum_{i=1}^{n} \binom{n}{i} p^i (1-p)^{n-i} \binom{m}{k-i} p^{k-i} (1-p)^{m-k+i} \\
&= \sum_{i=1}^{n} \binom{n}{i} \binom{m}{k-i} p^k (1-p)^{n+m-k} \\
&= p^k (1-p)^{n+m-k} \sum_{i=1}^{n} \binom{n}{i} \binom{m}{k-i} \\
&= p^k (1-p)^{n+m-k} \sum_{i=1}^{n} \binom{n+m}{k}
\end{aligned}$$

hence $Z \sim \textit{Binomial}(n+m, p)$.

The Pascal Distribution

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- Suppose we perform independent trials with probability of success p, until we observe m successes.
- Let X be the total number of trials. Note that $R_X = \{m, m+1, m+2, \dots\}$.
- What is the PMF of X?

$P(X = k) = P(A \cap B)$, where A is the event "m-1 successes in the first $k-1$ trials" and B is the event "success in the k-th trial". Since the trials are independent, we have

$$\begin{aligned} P(X = k) &= P(A)P(B) \\ &= \binom{k-1}{m-1} p^{m-1}(1-p)^{k-m} p \\ &= \binom{k-1}{m-1} p^{m}(1-p)^{k-m}. \end{aligned}$$

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A random variable X has the **Pascal distribution** with parameters m and p if

$$P(X = k) = \binom{k-1}{m-1} p^m (1-p)^{k-m}, \text{for } k = m, m+1, m+2, \ldots$$

and $P(X = k) = 0$ otherwise. Notation: $X \sim \textit{Pascal}(m, p)$.

- Special case ($m = 1$): $P(X = k) = p(1-p)^{k-1}$ for $k = 1, 2, \ldots$ and $P(X = k) = 0$ otherwise.
- Then X follows the **geometric distribution** with parameter p.
- Notation: $X \sim \textit{Geometric}(p)$.

The Pascal Distribution

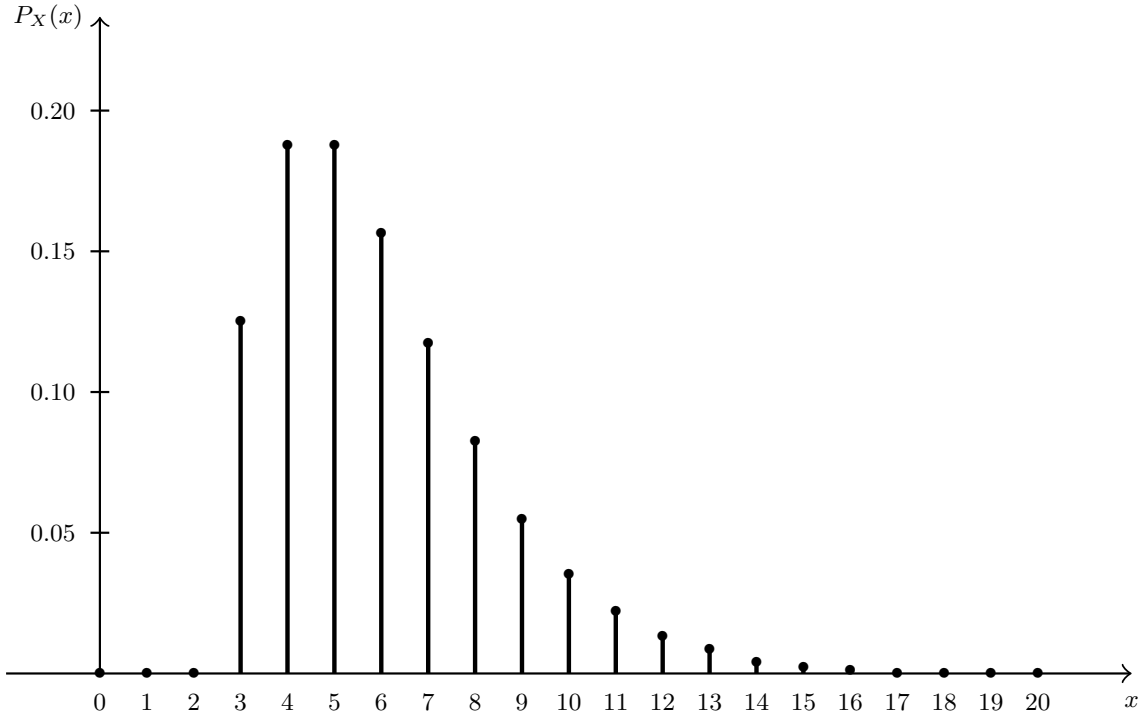

Figure: PMF of a $Pascal(3, 0.5)$ random variable

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The Poisson Distribution

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- Is used in counting the occurrences of events in an interval of time or space.
- Suppose we are counting the number of customers who visit a certain store between 1pm and 2pm.
- Based on data from previous days, we know that, on average, $\lambda = 15$ customers visit the store between 1pm and 2pm.
- The Poisson distribution with parameter $\lambda = 15$ could be a reasonable model for the number of customers X.

A random variable X has a **Poisson distribution** with parameter $\lambda > 0$ if

$$P_X(k) = \frac{e^{-\lambda}\lambda^k}{k!} \text{for } k = 0, 1, 2, \ldots$$

and $P_X(k) = 0$ otherwise. Notation: $X \sim \textit{Poisson}(\lambda)$.

Note that $\sum_{k=0}^{\infty} P_X(k) = \sum_{k=0}^{\infty} \frac{\lambda^k e^{-\lambda}}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{-\lambda} e^{\lambda} = 1.$

Poisson Distribution

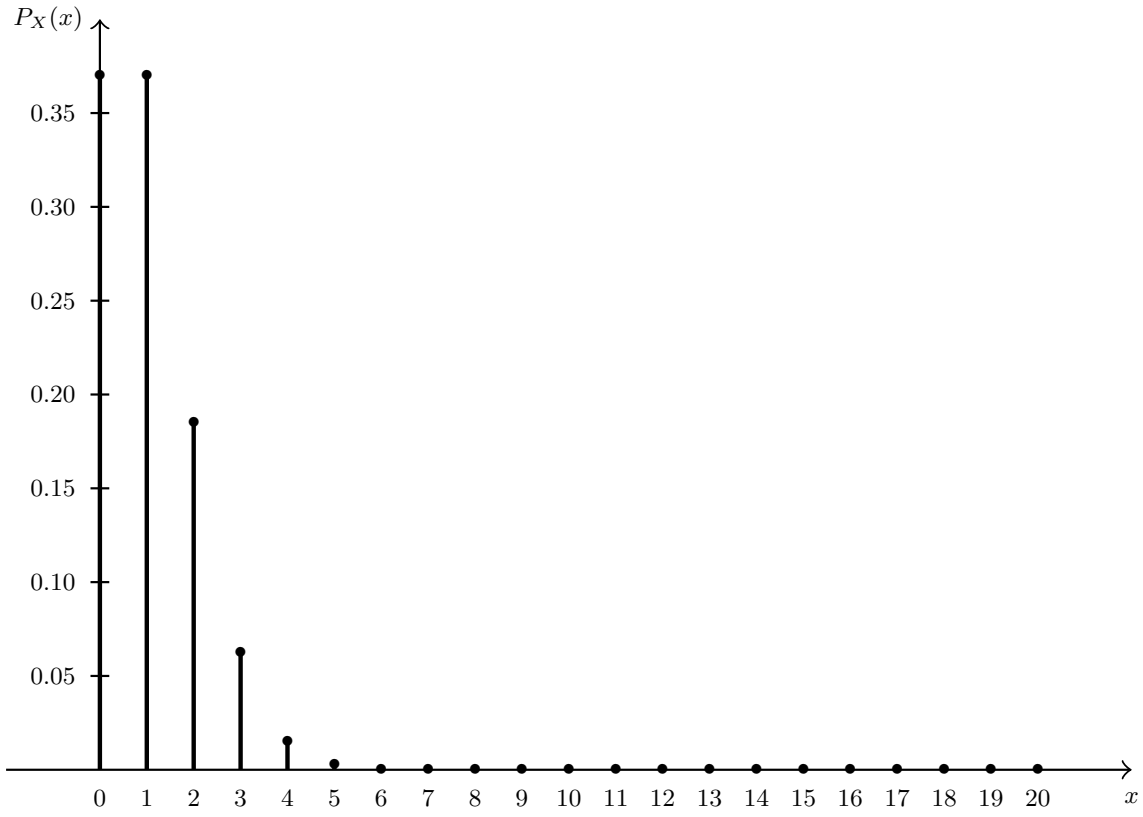

Figure: PMF of a $Poisson(1)$ random variable

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Poisson Distribution

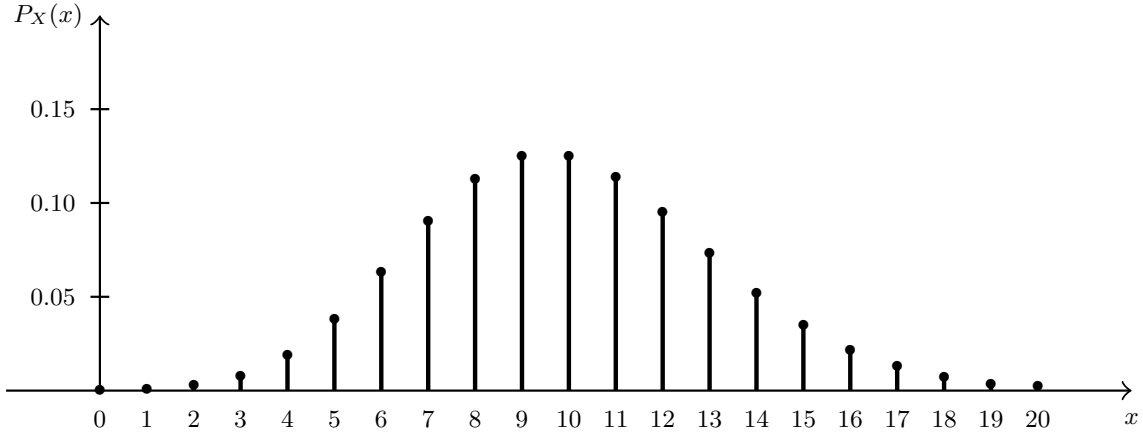

Figure: PMF of a $Poisson(10)$ random variable

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The Poisson Distribution

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Example (calculating probabilities)

The number of emails I receive per week can be modeled by a Poisson distribution with $\lambda = 0.2$ emails per minute. What is the probability that I receive no emails in an interval of 5 minutes?

Solution: Let X be the number of emails I receive in an interval of 5 minutes. Then $Y \sim \textit{Poisson}(5 \times 0.2) = \textit{Poisson}(1)$ and hence $P(Y = 0) = e^{-\lambda} = e^{-1} \approx 0.37$.

What is the probability that I receive more than 3 emails in an interval of 10 minutes?

Solution: Let Z be the number of emails I receive in an interval of 10 minutes. Then $Z \sim \textit{Poisson}(10 \times 0.2) = \textit{Poisson}(2)$ and hence

$$\begin{aligned} P(Z > 3) &= 1 - P(Z \leq 3) = 1 - P_Z(0) - P_Z(1) - P_Z(2) - P_Z(3) \\ &= 1 - e^{-\lambda}\left(1 + \lambda + \frac{\lambda^2}{2} + \frac{\lambda^3}{6}\right) = 1 - \frac{19}{3}e^{-2} \approx 0.14. \end{aligned}$$

The Poisson Distribution

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- Where does the formula for the Poisson distribution come from?
- It can be viewed as a limiting case of the binomial distribution:

Theorem. Let $X \sim Binomial(n, p = \lambda/n)$, where $\lambda > 0$ is fixed. Then

$$\lim_{n\to\infty} P_X(k) = \frac{e^{-\lambda}\lambda^k}{k!} \text{ for } k = 0, 1, 2, \ldots$$

- This is useful as the Poisson PMF is much easier to work with than the binomial PMF.

Exercises

Section 3.3: Exercises 7, 8, 9, 11

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The Cumulative Distribution Function (CDF)

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- The Cumulative distribution function (CDF) is an alternative way of specifying random variables.
- Theoretical arguments are often simpler when using the CDF instead of the PMF.

The **cumulative distribution function** (CDF) of a discrete random variable X is defined as

$$F_X(x) = P(X \leq x), \text{ for all } x \in \mathbb{R}$$

.

- F_X is defined not just on R_X but on the entire line $\mathbb{R}$.
- Note the use of $\leq$ (and not $<$).
- Note that $F_X(x) = \sum_{y \in R_y : y \leq x} P_X(y)$, where P_X is the PMF of X (hence the name "cumulative distribution").

The Cumulative Distribution Function (CDF)

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Example

We toss a coin two times and let X be the number of heads. Determine the CDF of X.

Solution: $X \sim$ *Binomial*$(2, 1/2)$ hence $P_X(0) = 1/4$, $P_X(1) = 1/2$, and $P_X(2) = 1/4$ (and otherwise zero). $F_X(x) = P(X \leq x)$. If $x < 0$, then $F_X(x) = 0$ and if $x > 2$, then $F_X(x) = 1$. Furthermore, if $0 \leq x < 1$, then $F_X(x) = P_X(0) = 1/4$. Similarly, if $1 \leq x < 2$, then $F_X(x) = P_X(0) + P_X(1) = 3/4$. Conclusion:

$$F_X(x) = \begin{cases} 0 & \text{for } x < 1 \\ 1/4 & \text{for } 0 \leq x < 1 \\ 3/4 & \text{for } 1 \leq x < 2 \\ 1 & \text{for } x \geq 2 \end{cases}$$

The Cumulative Distribution Function (CDF)

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- Let X be a discrete random variable with range $R_X = \{x_1, x_2, ...\}$ such that $x_1 < x_2 < ....$
- $F_X(x) = 0$ for $x < x_1$, is constant on $[x_k, x_{k+1})$, and at x_{k+1} it makes a jump of size $P(X_{k+1}) - P(x_k)$.

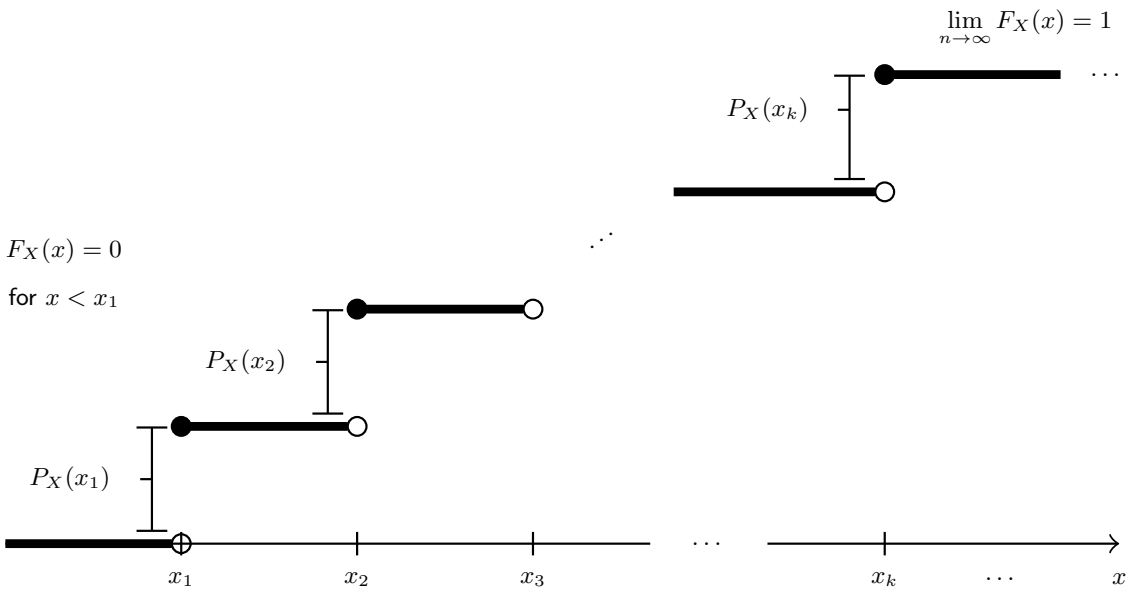

Figure: CDF of a discrete random variable with range $\{x_1, x_2, \dots\}$

The Cumulative Distribution Function (CDF)

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- The CDF is non-decreasing: If $x \leq y$ then $F_X(x) \leq F_X(y)$.
- $\lim_{x\to\infty} F_X(x) = 1$
- $F_X(x) = \sum_{x_k \leq x} P_X(x_k)$
- $P(a < X \leq b) = F_X(b) - F_X(a)$

This is because

$$\begin{aligned} P(a < X \leq b) &= \sum_{a < x_k \leq b} P(X = x_k) \\ &= \sum_{x_k \leq b} P(X = x_k) - \sum_{x_k \leq a} P(X = x_k) \\ &= F_X(b) - F_X(a). \end{aligned}$$

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Example

Let X be a discrete random variable with range $R_X = \{1, 2, ...\}$ and PMF $P_X(k) = 1/2^k$. Find and plot the CDF of X and use it to calculate $P(2 < X \leq 5)$.

Solution: $F_X(x) = 0$ for $x < 1$. For $1 \leq x < 2$ we have that $F_X(x) = F_X(1) = P_X(1) = 1/2$, for $2 \leq x < 3$ we have that $F_X(x) = F_X(2) = P_X(1) + P_X(2) = 1/2 + 1/4 = 3/4$ and, more generally, for $k \leq x < k+1$ we have that

$$F_X(x) = \frac{1}{2} + \frac{1}{2^2} + \cdots + \frac{1}{2^k} =^{5} \frac{1/2 - (1/2)^{k+1}}{1 - 1/2} = 1 - (1/2)^k.$$

Now:

$$P(2 < X \leq 5) = F_X(5) - F_X(2) = (1/2)^2 - (1/2)^5 = 7/32.$$

[5] Here we use that $\sum_{i=0}^{n} r^i = (1 - r^{n+1})/(1 - r)$ for $-1 < r < 1$ and hence $\sum_{i=1}^{n} r^i = (r - r^{n+1})/(1 - r)$.

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The **expected value** or **mean** of a discrete random variable X is

$$E(X) = \sum_{x \in R_X} x P(X = x)$$

Notation: $E(X) = EX = E[X] = \mu_X = \mu$.

Interpretation:

- Recall that X is the outcome of a random experiment.
- We perform the experiment N times, obtaining $x_1, \ldots, x_N \in \mathbb{R}$.
- Say we observe x_k, N_k times, hence $N_k/N \approx P_X(x_k)$ (for large N).
- The average of the observations is

$$\sum_{k=1}^{N} \frac{x_k N_k}{N} \approx \sum_{k=1}^{N} P_X(x_k) x_k = E(X).$$

Expectation

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Example (Bernoulli distribution)

Let $X \sim$ *Bernoulli*(p). Determine $E(X)$.

Solution: $R_X = \{0, 1\}$ and therefore

$$E(X) = \sum_{x \in R_X} x P_X(x) = 0 \times P_X(0) + 1 \times P_X(1) = 1 \times P_X(1) = p.$$

Example (throwing a die)

Throw a die and let $X \in \{1, 2, 3, 4, 5, 6\}$ be the outcome. Determine $E(X)$.

Solution:

$$E(X) = \sum_{x \in R_X} x P_X(x) = 1 \times \frac{1}{6} + 2 \times \frac{1}{6} + \cdots + 6 \times \frac{1}{6} = \frac{21}{6} = 3.5.$$

Expectation

- The expectation for different special distributions
  - $X \sim$ Bernoulli(p): $EX = p$.
  - $X \sim$ Geometric(p): $EX = \frac{1}{p}$.
  - $X \sim$ Poisson(λ): $EX = \lambda$.
  - $X \sim$ Binomial(n, p): $EX = np$.
  - $X \sim$ Pascal(m, p): $EX = \frac{m}{p}$.

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- Any function[6] $f(X_1, \ldots, X_n)$ of discrete random variables $X_1, \ldots, X_n$ is a discrete random variable.

Theorem. For discrete random variables $X_1, \ldots, X_n$:

$$E(X_1 + \cdots + X_n) = E(X_1) + \cdots E(X_n).$$

Example (application to the binomial distribution)

Let $X \sim$ *Binomial*(n, p). Determine $E(X)$.

Solution: $X = X_1 + \cdots + X_n$, where $X_i \sim$ *Bernoulli*(p) and so

$$E(X) = E(X_1) + \cdots + E(X_n) = np.$$

[6] Here $f : \mathbb{R}^n \to \mathbb{R}$.

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- Any function[7] $Y = g(X)$ of a discrete random variable X is a discrete random variable and hence has a range, PMF, CDF, and expected value.
- The range of Y is $R_Y = \{g(x) : x \in R_X\}$.
- The PMF of Y is

$$\begin{aligned} P_Y(y) &= P(Y = y) \\ &= P(g(X) = y) \\ &= P(\{x \in R_X : g(x) = y\}) \\ &= \sum_{x:g(x)=y} P_X(x). \end{aligned}$$

- P_Y is indeed a PMF:

$$\sum_{y \in R_Y} P_Y(y) = \sum_{y \in R_Y} \sum_{x:g(x)=y} P_X(x) = \sum_{x \in R_X} P_X(x) = 1.$$

[7] Here $g : \mathbb{R} \to \mathbb{R}$.

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Example

Let X be given by $P_X(k) = \frac{1}{5}$ for $k = -1, 0, 1, 2, 3$ and let $Y = 2|X|$.[8] Find the range and PMF of Y.

Solution: $Y = g(X)$ where $g(x) = 2|x|$ hence

$$R_Y = \{g(x) : x \in R_X\} = \{2|x| : x = -1, 0, 1, 2, 3\} = \{0, 2, 4, 6\}.$$

Furthermore:

$$P_Y(0) = P(Y = 0) = P(2|X| = 0) = P(X = 0) = 1/5$$
$$P_Y(2) = P(Y = 2) = P(2|X| = 2) = P(X = -1) + P(X = 1) = 2/5$$
$$P_Y(4) = P(Y = 4) = P(2|X| = 4) = P(X = 2) = 1/5$$
$$P_Y(6) = P(Y = 6) = P(2|X| = 6) = P(X = 3) = 1/5.$$

[8] Here $|a|$ denotes the absolute value of the number a.

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- Let X be a discrete random variable and let $Y = g(X)$. How to calculate $E(Y)$?
- We can use the definition: $E(Y) = \sum_{y \in R_Y} y P_Y(y)$, but this requires to first calculate the PMF of Y.
- Often it is easier to use LOTUS.

Law of the unconscious statistician (LOTUS). For any discrete random variable X and function g:

$$E(g(X)) = \sum_{x \in R_X} g(x) P_X(x).$$

Proof:

$$E(Y) = \sum_{y \in R_Y} y P_Y(y) = \sum_{y \in R_Y} y \sum_{x : g(x) = y} P_X(x) = \sum_{x \in R_X} g(x) P_X(x).$$

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Example (linearity of expectation)

Let X be a discrete random variable. Prove that $E(aX+b) = aE(X)+b$ for all $a, b \in \mathbb{R}$.

Solution: Let $g(x) = ax + b$. Using LOTUS we find:

$$\begin{aligned} E(aX+b) &= \sum_{x \in R_X} (ax+b) P_X(x) \\ &= \sum_{x \in R_X} axP_X(x) + \sum_{x \in R_X} bxP_X(x) \\ &= a \sum_{x \in R_X} xP_X(x) + b \sum_{x \in R_X} xP_X(x) \\ &= a \sum_{x \in R_X} xP_X(x) + b \\ &= aE(X) + b. \end{aligned}$$

Variance

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- Consider random variables X and Y with PMFs

$$P_X(x) = \begin{cases} 0.5 & \text{for } x = +100 \\ 0.5 & \text{for } x = -100 \\ 0 & \text{otherwise} \end{cases}$$

$$P_Y(y) = \begin{cases} 0.5 & \text{for } x = +1 \\ 0.5 & \text{for } x = -1 \\ 0 & \text{otherwise} \end{cases}$$

- $E(X) = E(Y) = 0$, but the two distributions are clearly not the same.
- The **variance** is a measure of how spread out the distribution of a random variable is.
- The variance of X is large, since its distribution is spread out, whereas the variance of Y is small, since its distribution is concentrated.

Variance

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The **variance** of a discrete random variable X with mean $E(X) = \mu$ is

$$\text{Var}(X) = E\left[(X-\mu)^2\right].$$

Notation: $\text{Var}(X) = \sigma_X^2 = \sigma^2$.

- The variance is the expected squared deviation of X from its mean.
- $\text{Var}(X) = E(g(X))$, where $g(x) = (x-\mu)^2$.

$$\text{Var}(X) = E(X^2) - E(X)^2$$

Proof:

$$\begin{aligned}\text{Var}(X) &= E\left[(X-\mu)^2\right] = E\left[X^2 - 2\mu X + \mu^2\right] \\ &= E(X^2) - 2\mu E(X) + \mu^2 = E(X^2) - 2\mu^2 + \mu^2 \\ &= E[X^2] - \mu^2 = E[X^2] - E(X)^2.\end{aligned}$$

Standard Deviation and Useful Results

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The **standard deviation** of a discrete random variable X is

$$\mathsf{SD}(X) = \sigma_X = \sqrt{\mathsf{Var}(X)}.$$

- The standard deviation has the same units (i.e. cm or sec) as X.

$$\mathsf{Var}(aX + b) = a^2\mathsf{Var}(X)$$

For any **independent** discrete random variables $X_1, \ldots, X_n$:

$$\mathsf{Var}(X_1 + \cdots + X_n) = \mathsf{Var}(X_1) + \cdots + \mathsf{Var}(X_n).$$

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Example (variance of binomial distribution)

Let $X \sim$ *Binomial*(n, p). Determine Var(X).

Solution: $X = X_1 + \cdots + X_n$, with $X_1, \ldots, X_n$ independent and $X_i \sim$ *Bernoulli*(p), hence Var$(X) = n$Var(X_i). Furthermore,

$$\text{Var}(X_i) = E(X_i^2) - E(X_i)^2 = 0^2 \times (1-p) + 1^2 \times p - p^2 = p(1-p)$$

and therefore Var$(X) = np(1-p)$.

Exercises

Section 3.3: Exercises 12, 13, 14, 15, 17, 19, 22

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